

29 May 2026 Urchin Cage Set-Up

Harkers Island Bridge

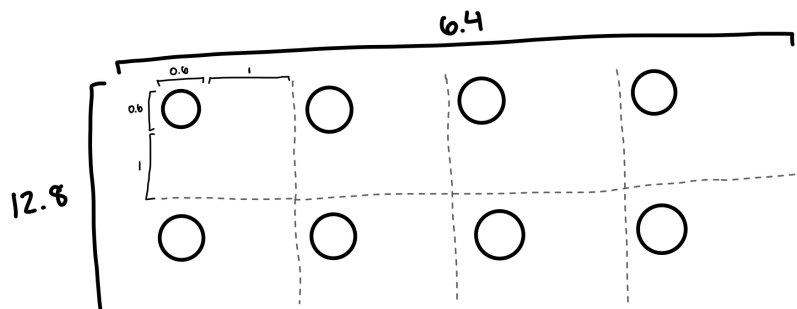
Question: What is the impact of urchin density on seagrass beds?

Goals:

- Set up cages
- Take initial measurements

Cage set up

Cages are 0.0625m^2 circles that are 30cm tall.



Spaced 1 meter away from each other. This means the total space allotted to each plot is $1.6 \times 1.6\text{m}$. They will be set up in 4 rows of 8.

Remove any macro algae from the cage area during set up.

Initial measurements

All measurements:

- Percent cover
 - Halodule
 - Zostera
- Shoot density
 - Whole plot
 - $10 \times 10\text{cm}$ quadrat
- Algal cover
 - Remove during set up
 - Percent cover thereafter
- Shoot height
- Epiphyte coverage

Percent cover

To measure percent coverage of seagrass do both Zostera and Halodule separately

Shoot density

Count the number of shoots in a given area, or just in a 10x10 cm area, make the call in the field.

Algal cover

Estimate the percent cover of macroalgae within the plot

Shoot height + epiphyte coverage

Measure shoot height of 10 random blades. Also estimate percent epiphyte coverage.

Monitoring Plan

- If any dead urchins are found, remove from plot and replace with healthy urchin.
- Ensure cages are sound and repair any issues.

Materials:

- Cages (32)
- Tags on cages
- Urchins (65) in cooler
- Lawn staples (100+)
- Datasheet + clipboard + pencil(s)
- 10cm quadrat
- Flags
- Transect line
- Measuring stick
- Extra zip ties
- Scissors
- Bucket (for collection)

1 1 urchin	2 2 urchin	3 0 urchin	4 2 urchin
8 2 urchin	7 0 urchin	6 4 urchin	5 2 urchin
9 4 urchin	10 4 urchin	11 1 urchin	12 1 urchin
16 0 urchin	15 4 urchin	14 2 urchin	13 0 urchin
17 2 urchin	18 0 urchin	19 1 urchin	20 0 urchin
24 4 urchin	23 1 urchin	22 0 urchin	21 4 urchin
25 0 urchin	26 1 urchin	27 4 urchin	28 4 urchin
32 1 urchin	31 1 urchin	30 4 urchin	29 2 urchin